

Møller Polarimeter Aperture Scintillator Detector

18 March 2013

This is a short description of a new segmented aperture scintillators detector for the Møller detector. New aperture detector consists of 4 segments each (left and right). The segments have a different length and width. The longest one is located outside of the detector center. Other three segments are located on the site closer to the detector center. The sketches of the aperture detector are shown here (pdf format) and here (pdf format). Photo of the aperture detector in assembly is shown [here](#). The Hamamatsu R4124 PMTs are used for all 8 channels of the aperture detector.

Below is a numbering of the aperture scintillator segments:

- the longest one is number #1
- upper segment is number #2
- middle segment is number #3
- lower segment is number #4

Below is the aperture scintillators connection to the signal panel in the Møller old DAQ rack.

Signal Name	Cable #	Panel Connector #
Ap_Left_#1	033	33
Ap_Left_#2	034	34
Ap_Left_#3	035	35
Ap_Left_#4	036	36
Ap_Left_#5	037	37
Ap_Left_#6	038	38
Ap_Left_#7	039	39
Ap_Left_#8	040	41

High voltage cables for aperture scintillators:

Signal Name	Cable #
Ap_Left_#1	013
Ap_Left_#2	014
Ap_Left_#3	015
Ap_Left_#4	016
Ap_Left_#5	017
Ap_Left_#6	018
Ap_Left_#7	019
Ap_Left_#8	020

High voltage channels (slot #6) for aperture scintillators:

Signal Name	HV channel #
Ap_Left_#1	00
Ap_Left_#2	01
Ap_Left_#3	02
Ap_Left_#4	03
Ap_Left_#5	04
Ap_Left_#6	05
Ap_Left_#7	06
Ap_Left_#8	07

To control HV settings for aperture (slot #6) and calorimeter (slot #5) channels one should login to HV mainframe using command:

```
telnet hatsv5 2003
```

and then start 'vt100' program.

Aperture scint. signals from panel connectors going to the amplifier PS776 (delay is 4ns): panel connectors #33–#40 to PS776 channels 1-8. From the outputs (#1-8) of this amplifier signals are connected to inputs of the FADC.

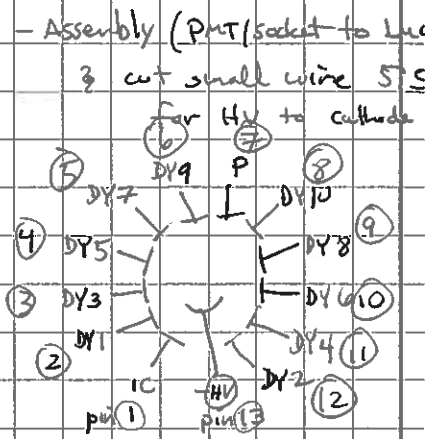
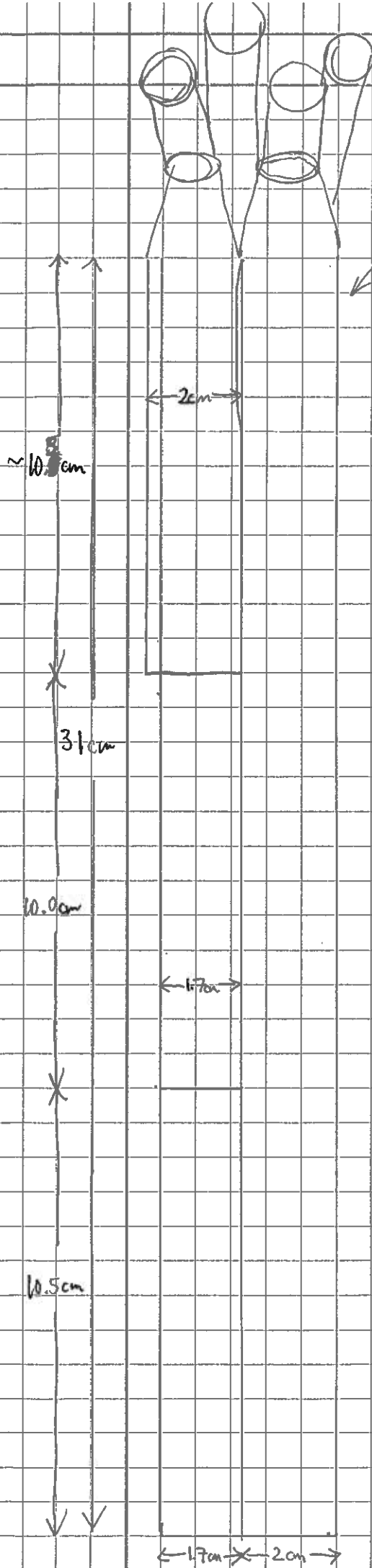
The delay between aperture scintillators and calorimeter signals on the panel connectors is about **~10-12ns** (aperture scint. signal appears early).

For old DAQ we take signals also from output of the amplifier and connecting its to the discriminator PS706 with threshold presets to -200mV and then send its to OR module PS740 to have two sums of four left and four right aperture signals, that we can use without of modification old DAQ.

Figures

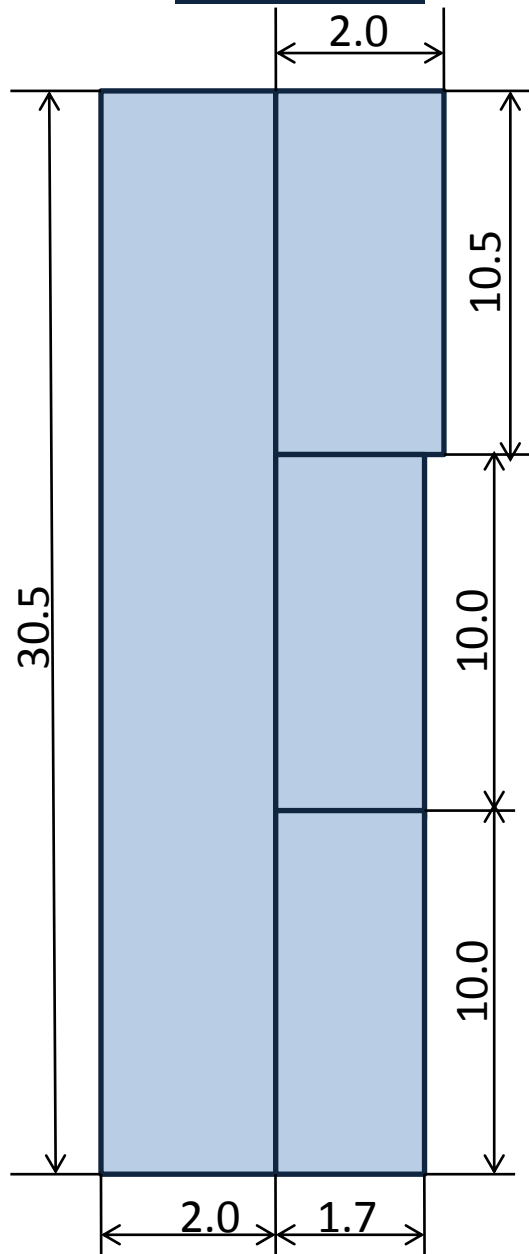
Photo of the aperture detector in assembly



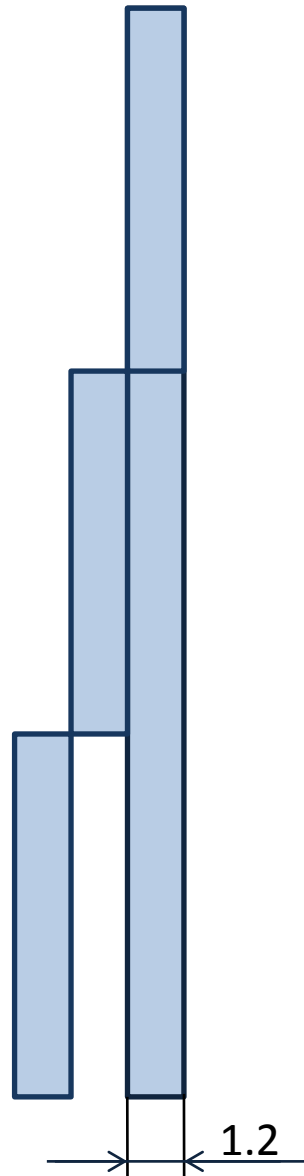


Hall A Møller Polarimeter Aperture Detector (Left)

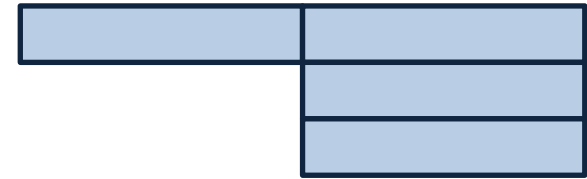
Front view



Side view



Top view



Sketch not in scale
Dimensions in cm

Fast Time Response (Especially Narrow T.T.S.)
13mm (1/2 Inch) Diameter, 10-Stage, Bialkali Photocathode, Head-On Type

GENERAL

Parameter		Description/Value	Unit
Spectral Response		300 to 650	nm
Wavelength of Maximum Response		420	nm
Photocathode	Material	Bialkali	—
	Minimum Effective Area	10	mm dia.
Window Material		Borosilicate glass	—
Dynode	Structure	Linear focused	—
	Number of Stages	10	—
Base		13-pin glass base	—
Weight		Approx. 7	g
Suitable Socket		E678-13A (supplied)	—

MAXIMUM RATINGS (Absolute Maximum Values)

Parameter		Value	Unit
Supply Voltage	Between Anode and Cathode	1250	Vdc
	Between Anode and Last Dynode	350	Vdc
Average Anode Current		0.03	mA
Ambient Temperature		-80 to +50	°C

CHARACTERISTICS (at 25°C)

Parameter		Min.	Typ.	Max.	Unit
Cathode Sensitivity	Luminous (2856K)	40	95	—	μA/lm
	Radiant at 420nm	—	76	—	mA/W
	Blue (CS 5-58 filter)	—	9.5	—	μA/lm-b
Anode Sensitivity	Luminous (2856K)	30	100	—	A/lm
Gain		—	1.1×10^6	—	—
Anode Dark Current (after 30min. storage in darkness)		—	1	15	nA
Time Response	Anode Pulse Rise Time	—	1.1	—	ns
	Electron Transit Time	—	12	—	ns
	Transit Time Spread (FWHM)	—	0.5	—	ns

VOLTAGE DISTRIBUTION RATIO AND SUPPLY VOLTAGE

Electrodes	K	Dy1	Dy2	Dy3	Dy4	Dy5	Dy6	Dy7	Dy8	Dy9	Dy10	P
Ratio	3	1	1.5	1	1	1	1	1	1	1	1	1

Supply Voltage: 1000Vdc, K: Cathode, Dy: Dynode, P: Anode

PHOTOMULTIPLIER TUBE R4124

Figure 1: Typical Spectral Response

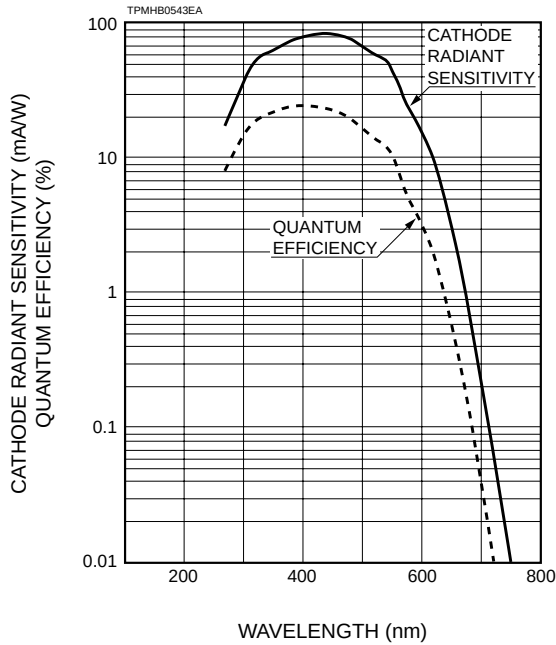


Figure 2: Typical Gain

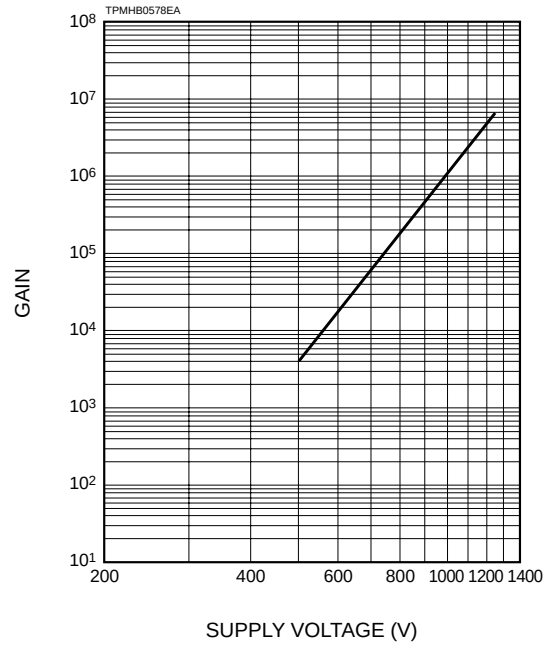
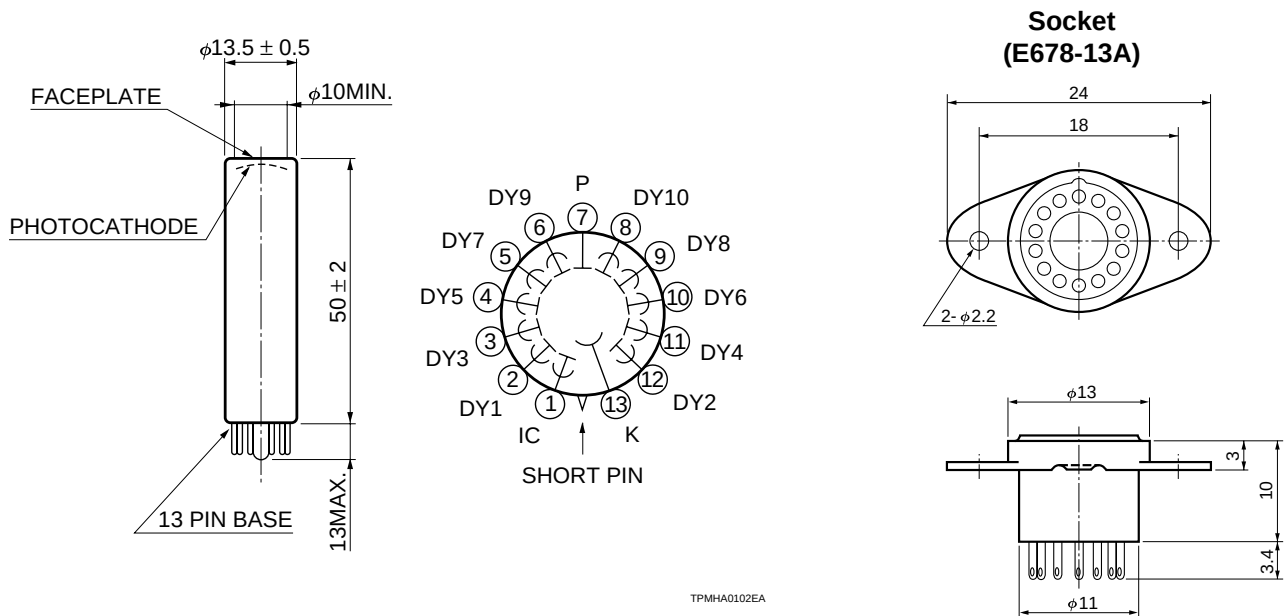


Figure 3: Dimensional Outline and Basing Diagram (Unit: mm)



TPMHA0102EA

TACCA0005EA

HAMAMATSU

HAMAMATSU PHOTONICS K.K., Electoron Tube Center

314-5, Shimokanzo, Toyooka-village, Iwata-gun, Shizuoka-ken, 438-0193, Japan, Telephone: (81)539/62-5248, Fax: (81)539/62-2205

U.S.A.: Hamamatsu Corporation: 360 Foothill Road, P. O. Box 6910, Bridgewater, N.J. 08807-0910, U.S.A., Telephone: (1)908-231-0960, Fax: (1)908-231-1218

Germany: Hamamatsu Photonics Deutschland GmbH: Arzbergerstr. 10, D-82211 Herrsching am Ammersee, Germany, Telephone: (49)8152-375-0, Fax: (49)8152-2658

France: Hamamatsu Photonics France S.A.R.L.: 8, Rue du Saule Trappu, Parc du Moulin de Massy, 91882 Massy Cedex, France, Telephone: (33)1 69 53 71 00, Fax: (33)1 69 53 71 10

United Kingdom: Hamamatsu Photonics UK Limited: Lough Point, 2 Gladbeck Way, Windmill Hill, Enfield, Middlesex EN2 7JA, United Kingdom, Telephone: (44)181-367-3560, Fax: (44)181-367-6384

North Europe: Hamamatsu Photonics Norden AB: F  r  gatan 7, S-164-40 Kista Sweden, Telephone: (46)8-703-29-50, Fax: (46)8-750-58-95

Italy: Hamamatsu Photonics Italia: S.R.L.: Strada della Moia, 1/E, 20020 Arese, (Milano), Italy, Telephone: (39)02-935 81 733, Fax: (39)02-935 81 741

TPMH1233E01
SEPT. 1998